

ANALYSTLAB AFRICA

Experience Lab Internship Programme

Week 5: Exploratory Analysis, KPI Development & Business Insights

Data Analytics Track — Week 5 Project Summary

Field	Detail
Project Title	Improving Patient Appointment Attendance and Healthcare Support Using Data and AI
Prepared by	Okemgbo Oluebube
Track	Data Analytics
Week	Week 5

1. What I Planned to Accomplish

Building on the Week 4 problem understanding phase, Week 5 was planned to move from problem definition into actual analysis: preparing the dataset for calculation, conducting exploratory data analysis across the variables identified in Week 4, calculating the five proposed KPIs, building an initial analytical dashboard, and producing at least five meaningful business insights with recommendations.

2. What I Actually Completed

- Prepared the dataset for analysis in Excel and Power BI/Power Query, building on the Week 4 data quality assessment, without altering the original source file.
- Conducted exploratory analysis across eleven dimensions: monthly trends, previous no-show history, age group, gender, reminder status, reminder channel, booking lead time, appointment day, distance to clinic, waiting time, and appointment type.
- Calculated all five proposed KPIs — Total Appointments, Attendance Rate, No-Show Rate, Cancellation Rate, and Previous No-Show Rate — with each defined, interpreted, and linked back to its business question.
- Built an initial analytical dashboard in Power BI summarising the KPIs and key exploratory relationships.
- Documented seven key findings with corresponding business recommendations, a cross-track collaboration point with the Data Science track, and a full review of the Week 4 assumptions, limitations, risks, and dependencies.

3. Key Findings / Development Outcomes

The headline result is that 51.15% of non-cancelled appointments were No-Shows, against an Attendance Rate of 48.85% and a Cancellation Rate of 5.26%. 41.58% of patients have a history of at least one previous missed appointment, and this history is strongly associated with current no-show behaviour.

The strongest observed predictors of no-show were previous no-show history, reminder status and channel (SMS best, WhatsApp worst), booking lead time (longer lead time, higher risk), and appointment day (highest on Monday, Sunday, and Wednesday; lowest on Friday). Age, gender, and waiting time showed weak or inconsistent relationships with no-show behaviour, and distance to clinic showed a broadly increasing trend with one unexplained anomaly at 40–50km.

4. Major Challenges Encountered

- An early DAX measure for No-Show Rate referenced a mismatched measure name, which was a reminder to verify measure names carefully when chaining calculations together.

- The 40–50km distance band did not follow the otherwise increasing no-show trend, and the cause could not be determined from the available data alone.
- The exact peak no-show rate associated with the longest booking lead times was not fully captured during this round of analysis.

5. Important Decisions Made and Why

- Decided to calculate Attendance Rate and No-Show Rate as a percentage of non-cancelled appointments only (Attended + No-Show), consistent with the KPI definitions proposed in Week 4, rather than as a percentage of all 5,000 appointments. This is why the Week 5 No-Show Rate (51.15%) and Attendance Rate (48.85%) read slightly higher than the equivalent Week 4 figures (48.46% and 46.28%, calculated as a share of all appointments including cancellations) — the underlying counts are unchanged; only the denominator differs, and the Week 5 approach better isolates genuine attendance behaviour from cancellations.
- Decided to flag the 40–50km distance anomaly explicitly rather than smooth it into the overall distance trend, to avoid overstating a clean relationship that the data does not fully support.
- Decided to lower the priority of waiting_time_minutes as an explanatory factor, since no consistent relationship with no-show rate was found, in contrast to its Week 4 classification as a potential patient-experience factor.

6. Changes to the Week 4 Approach

No structural changes were made to the Week 4 problem definition, variables, or business questions. The main refinement was applying the non-cancelled denominator consistently when calculating Attendance Rate and No-Show Rate, exactly as defined in the Week 4 KPI table, which produced the slightly higher percentages noted in Section 5 above.

7. Cross-Track Collaboration Completed

Shared the KPI results and strongest observed predictors of no-show (previous no-show history, reminder status and channel, booking lead time, appointment day, appointment type) with the Data Science track, to support feature selection for their proposed no-show prediction model. Also flagged distance_to_clinic_km and waiting_time_minutes as variables needing closer inspection due to inconsistent or anomalous relationships with the outcome. Full details are documented in the main Week 5 Analytics Report.

8. Remaining Work

- Insert final Power BI dashboard screenshots into the main analytics report.
- Verify the exact definition of waiting_time_minutes against the HealthConnect Data Dictionary.
- Investigate the cause of the 40–50km distance anomaly.
- Confirm the exact peak no-show rate associated with the longest booking lead times.

9. Proposed Focus for Week 6

The proposed focus for Week 6 is to deepen the analysis where gaps remain — resolving the distance anomaly and the waiting-time definition — and to begin translating the Week 5 findings and recommendations into a more refined, presentation-ready dashboard, in preparation for later stages of the HealthConnect Experience Lab.